Q1. Data Preprocessing in Retail Analytics

Below is a complete, step-by-step solution that you can write for your CO2 – Assessment Tool 3.

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

CO2 – Assessment Tool 3

Course: Data Warehousing and Data Mining

Q1. Data Preprocessing in Retail Analytics

Given Dataset

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C8	42	Female	300	45	High

A. Handling Missing Values

(i) Identify Missing Values

Column	Missing Value
Age	C2
Monthly_Spend	C3

Column	Missing Value
Purchase_Frequency	C5

(ii) Apply Two Imputation Techniques

Method 1: Median Imputation

Age

Available values

23, 31, 105, 27, 40, 36, 42

Arrange in ascending order

23, 27, 31, 36, 40, 42, 105

Median = 36

Therefore,

Age(C2) = 36

Monthly Spend

Available values

20, 35, 500, 25, 45, 38, 300

Sorted

20, 25, 35, 38, 45, 300, 500

Median = 38

Therefore,

Monthly Spend(C3) = 38

Purchase Frequency

Available values

5,8,10,50,12,9,45

Sorted

5,8,9,10,12,45,50

Median = 10

Therefore,

Purchase Frequency(C5)=10

Method 2: k-NN Imputation

k-NN finds the nearest customers based on similar features.

Example

Customer C2

Nearest customers

- **C6**
- **C7**
- **C8**

Average Age

=(40+36+42)/3

=39.33

≈ 39 years

Similarly,

Monthly Spend(C3)

Nearest customers

35,38,45

Average

=(35+38+45)/3

=39.33

≈ 39

Purchase Frequency(C5)

Nearest neighbours

8,9,12

Average

=(8+9+12)/3

=9.67

≈ 10

(iii) Justification

Median Imputation

- **Easy to calculate.**
- **Not affected by extreme values.**
- **Suitable because Age and Monthly Spend contain outliers.**

k-NN Imputation

- **Uses similar records.**
- **Produces more realistic values.**
- **Maintains relationships among attributes.**

B. Outlier Detection and Treatment

(i) Detect Outliers Using Z-score

Formula

$$Z = \frac{x - \mu}{\sigma}$$

where

- **x = observation**
- **μ = mean**
- **σ = standard deviation**

Age

Values

23,36,31,105,27,40,36,42

Mean

$$\mu = \frac{340}{8} = 42.5$$

Standard deviation

≈24.35

Age =105

$$Z = \frac{105 - 42.5}{24.35} = 2.57$$

Since

|Z| > 2

Age 105 is an outlier.

Monthly Spend

Values

20,35,38,500,25,45,38,300

Mean

$$= \frac{1001}{8} = 125.13$$

Standard deviation

≈173

Spend=500

$$Z = \frac{500 - 125.13}{173} = 2.17$$

Outlier

Spend=300

$$Z = \frac{300 - 125.13}{173} = 1.01$$

Not considered an extreme outlier.

(ii) Records Containing Outliers

Customer	Outlier
C4	Age =105
C4	Monthly Spend =500

(iii) Treatment

Capping

Replace Age

105

↓

Maximum acceptable age

60

Replace Spend

500

↓

Maximum spend

100

Removal

If the record is entered incorrectly,

Remove C4 completely.

Transformation

Use

Log Transformation

to reduce skewness.

Justification

Capping preserves records while reducing the influence of extreme values.

Transformation reduces skewness without deleting information.

Removal is suitable only when the value is clearly erroneous.

C. Categorical Data Standardization

(i) Identify Inconsistencies

Original values

M

Male

male

F

Female

FEMALE

Different spellings represent the same category.

(ii) Standardize Gender

Original	Standardized
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Final Cleaned Dataset

Customer	Age	Gender	Monthly Spend	Purchase Frequency	Loyalty
C1	23	Male	20	5	Low
C2	36 (Median)	Male	35	8	Medium
C3	31	Female	38 (Median)	10	High
C4	60 (Capped)	Female	100 (Capped)	50	High
C5	27	Male	25	10 (Median)	Low
C6	40	Female	45	12	Medium
C7	36	Male	38	9	Low
C8	42	Female	300	45	High

Q2. Covariance & Data Reduction in E-commerce Analytics

Given Dataset

Cust_ID	Age	Income (₹K)	Purchase Value (₹K)	Credit Score	Orders	Avg Balance (₹K)	EMI (₹)	Customer Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

A. Covariance Analysis

(i) Compute Covariance between Income and Purchase Value

Formula

$$\text{Cov}(X,Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

Where

- X=Income
- Y=Purchase Value

Step 1: Calculate Mean

Mean Income

$$\bar{X} = \frac{40+60+85+120+45+75+150+70}{8} = \frac{645}{8} = 80.625$$

Mean Purchase Value

$$\bar{Y} = \frac{100+150+210+320+110+180+360+160}{8} = \frac{1590}{8} = 198.75$$

Step 2: Compute Deviations

Income	Purchase	Xi−X̄	Yi−Ȳ	(Xi−X̄)(Yi−Ȳ)
40	100	-40.625	-98.75	4011.72
60	150	-20.625	-48.75	1005.47

Income	Purchase	Xi–X̄	Yi–Ȳ	(Xi–X̄)(Yi–Ȳ)
85	210	4.375	11.25	49.22
120	320	39.375	121.25	4774.22
45	110	-35.625	-88.75	3161.72
75	180	-5.625	-18.75	105.47
150	360	69.375	161.25	11187.89
70	160	-10.625	-38.75	411.72

Step 3: Sum

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 24707.43$$

Step 4: Covariance

$$\text{Cov}(IncomePurchase) = \frac{24707.43}{7} = 3529.63$$

Covariance ≈ 3529.63

(ii) Interpretation

The covariance value is positive (+3529.63).

This means:

- **Income increases.**
- **Purchase Value also increases.**

Therefore,

Higher-income customers generally spend more money.

(iii) Importance of Covariance

Covariance helps to:

- **Measure the relationship between two variables.**
- **Identify whether variables increase or decrease together.**
- **Detect highly related features.**

- Reduce redundant features before model building.

B. Feature Selection

(i) Identify Redundant Attributes

Some attributes are highly related.

Attribute 1	Attribute 2	Reason
Income	Purchase Value	Higher income usually increases purchases
Income	Average Balance	Customers with higher income maintain higher balances
Purchase Value	Orders	More orders usually increase purchase value
Income	EMI	Higher income customers often have higher EMI payments

These features provide similar information.

(ii) Optimal Feature Selection

Selected Features

- Age
- Income
- Credit Score
- Orders
- Average Balance

Removed Features

- Purchase Value
- EMI

(iii) Justification

Income already represents purchasing power.

Average Balance represents financial stability.

Credit Score measures customer reliability.

Orders indicate buying behaviour.

Removing Purchase Value and EMI reduces redundancy and computational cost.

C. Data Transformation

(i) Standardization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

where

- **X = Value**
- **μ = Mean**
- **σ = Standard Deviation**

Example

Income

Mean = 80.625

Standard deviation ≈35.93

For U1

Income=40

$$Z = \frac{40 - 80.625}{35.93} = -1.13$$

For U2

Income=60

$$Z = \frac{60 - 80.625}{35.93} = -0.57$$

For U3

Income=85

$$Z = \frac{85 - 80.625}{35.93} = 0.12$$

The same process is applied to:

- **Age**
- **Income**
- **Purchase Value**

- **Credit Score**
- **Orders**
- **Average Balance**
- **EMI**

Cust_ID and Customer_Class are not standardized because they are identifiers and target labels.

(ii) Why Standardization is Required

Standardization is important because:

- **All features are converted to the same scale.**
- **Large values do not dominate smaller values.**
- **Improves model accuracy.**
- **Speeds up machine learning algorithms.**
- **Helps distance-based algorithms perform better.**

Final Selected Features

Feature	Status
Age	Selected
Income	Selected
Credit Score	Selected
Orders	Selected
Average Balance	Selected
Purchase Value	Removed (Highly Correlated)
EMI	Removed (Redundant)

Feature Status

Q3. Data Transformation & Discretization in Education Analytics

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Given Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

A. Equal-Width Binning

Equal-width binning divides the data into bins of equal interval size.

(i) Apply 3 bins for Study_Hours

Step 1: Find Minimum and Maximum

Minimum = 2

Maximum = 11

Step 2: Calculate Bin Width

$$\text{Bin Width} = \frac{\text{Maximum} - \text{Minimum}}{\text{Number of Bins}}$$

$$= \frac{11 - 2}{3} = \frac{9}{3} = 3$$

Bin Width = 3

Step 3: Define Intervals

Bin Interval

Bin 1 2 – 5

Bin 2 5 – 8

Bin	Interval
Bin 3	8 – 11

Step 4: Assign Values

Student	Study_Hours	Bin
S1	2	Bin 1
S2	3	Bin 1
S3	4	Bin 1
S4	5	Bin 1
S5	6	Bin 2
S6	7	Bin 2
S7	8	Bin 2
S8	9	Bin 3
S9	10	Bin 3
S10	11	Bin 3

Equal-Width Binning for Test Score

Minimum = 40

Maximum =100

Bin Width

$$= \frac{100 - 40}{3} = \frac{60}{3} = 20$$

Bin Width = 20

Intervals

Bin	Interval
Bin 1	40 – 60

Bin	Interval
Bin 2	60 – 80
Bin 3	80 –100

Assign Values

Student	Score	Bin
S1	40	Bin1
S2	50	Bin1
S3	55	Bin1
S4	65	Bin2
S5	75	Bin2
S6	85	Bin3
S7	90	Bin3
S8	95	Bin3
S9	98	Bin3
S10	100	Bin3

B. Equal-Frequency Binning

Equal-frequency binning divides the data so that each bin contains nearly the same number of records.

Total Records = 10

Number of Bins = 3

Each bin contains approximately 3–4 records.

Study Hours

Bin	Values
Bin 1	2,3,4,5
Bin 2	6,7,8

Bin	Values
Bin 3	9,10,11

Test Score

Bin	Values
Bin 1	40,50,55,65
Bin 2	75,85,90
Bin 3	95,98,100

Student Assignment

Student	Study Hours Bin	Test Score Bin
S1	Bin1	Bin1
S2	Bin1	Bin1
S3	Bin1	Bin1
S4	Bin1	Bin1
S5	Bin2	Bin2
S6	Bin2	Bin2
S7	Bin2	Bin2
S8	Bin3	Bin3
S9	Bin3	Bin3
S10	Bin3	Bin3

C. Concept Hierarchy Construction

Concept hierarchy converts numerical values into meaningful categories.

Study Hours

Hours	Category
2–5	Low

Hours	Category
6–8	Medium
9–11	High

Test Score

Score	Category
40–60	Poor
61–80	Average
81–100	Excellent

Student	Study Hours	Category	Test Score	Category
S1	2	Low	40	Poor
S2	3	Low	50	Poor
S3	4	Low	55	Poor
S4	5	Low	65	Average
S5	6	Medium	75	Average
S6	7	Medium	85	Excellent
S7	8	Medium	90	Excellent
S8	9	High	95	Excellent
S9	10	High	98	Excellent
S10	11	High	100	Excellent

Q4. Use the two methods below to normalize the following data

Given Data

200, 300, 400, 600, 1000

(a) Min-Max Normalization

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Given

- **Minimum = 200**
- **Maximum = 1000**

Range

$$\mathbf{1000 - 200 = 800}$$

Calculations

For 200

$$\frac{\mathbf{200 - 200}}{\mathbf{800}} = \mathbf{0}$$

For 300

$$\frac{\mathbf{300 - 200}}{\mathbf{800}} = \frac{\mathbf{100}}{\mathbf{800}} = \mathbf{0.125}$$

For 400

$$\frac{\mathbf{400 - 200}}{\mathbf{800}} = \frac{\mathbf{200}}{\mathbf{800}} = \mathbf{0.25}$$

For 600

$$\frac{\mathbf{600 - 200}}{\mathbf{800}} = \frac{\mathbf{400}}{\mathbf{800}} = \mathbf{0.50}$$

For 1000

$$\frac{\mathbf{1000 - 200}}{\mathbf{800}} = \frac{\mathbf{800}}{\mathbf{800}} = \mathbf{1}$$

Final Answer

Original Value	Normalized Value
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-Score Normalization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Step 1: Mean

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

Step 2: Standard Deviation

X	X−μ	(X−μ)²
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000

$$\sum (X - \mu)^2 = 400000$$

Sample Standard Deviation

$$\sigma = \sqrt{\frac{400000}{4}} = \sqrt{100000} = 316.23$$

Step 3: Z-score

For 200

$$Z = \frac{200 - 500}{316.23} = -0.95$$

For 300

$$= -0.63$$

For 400

$$= -0.32$$

For 600

$$= 0.32$$

For 1000

$$= 1.58$$

Final Answer

Original Value	Z-Score
200	-0.95
300	-0.63
400	-0.32
600	0.32
1000	1.58

Q5. Data Transformation in Online Shopping Behavior Analysis

Given Dataset

Customer	Visit Duration (Seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800

Customer	Visit Duration (Seconds)
C10	3600

A. Equal-Frequency Binning

(i) Divide into 3 Bins

Total Records = 10

Each bin contains approximately 3–4 records.

Bin Assignment

Bin	Values
Bin 1	15,30,45,60
Bin 2	120,300,600
Bin 3	900,1800,3600

Customer-wise

Customer	Visit Duration	Bin
C1	15	Bin1
C2	30	Bin1
C3	45	Bin1
C4	60	Bin1
C5	120	Bin2
C6	300	Bin2
C7	600	Bin2
C8	900	Bin3
C9	1800	Bin3
C10	3600	Bin3

(iii) Analysis

- **Bin 1 groups short visits.**
- **Bin 2 groups medium-duration visits.**
- **Bin 3 groups long visits.**
- **This method keeps nearly the same number of records in each group, making it useful for clustering.**

B. Z-Score Normalization

(i) Mean

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10} = \frac{7470}{10} = 747$$

Standard Deviation

$$\sigma \approx 1125.76$$

(ii) Z-Score Calculation

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Visit Duration	Z-Score
15	-0.65
30	-0.64
45	-0.62
60	-0.61
120	-0.56
300	-0.40
600	-0.13
900	0.14
1800	0.94

Visit Duration	Z-Score
3600	2.53

(iii) Discussion

- **Very large values (1800 and 3600) receive high positive Z-scores.**
- **Z-score identifies extreme values (outliers) effectively.**
- **Suitable for statistical analysis and machine learning.**

C. Min-Max Normalization

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Minimum = 15

Maximum = 3600

Range =

$$3600 - 15 = 3585$$

Calculations

Visit Duration	Normalized Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498

Visit Duration	Normalized Value
3600	1.000

(iii) Comparison

Feature	Z-Score Normalization	Min-Max Normalization
Range	No fixed range	Values between 0 and 1
Mean	Centered around 0	Not centered
Outliers	Clearly highlights outliers	Sensitive to outliers because extreme values determine the scale
Best Use	Statistical analysis and ML algorithms	Neural networks, clustering, and visualization